

Conceptual model workshop

Objectives

Create an Entity-Relationship Diagram that captures the relationships and entities involved in managing course registrations, class scheduling, and instructor assignments at a university.

ERD for a University Management System

Scenario:

You are tasked with designing a database for a university to manage its academic operations efficiently. The database should keep track of students, courses, class sessions, seats in each class, instructors, and professors. Each entity and their relationship with others should clearly depict how the academic processes are interlinked.

Requirements(1)

Students: Each student should have a unique identifier, a name, and an address.

- Students register for various courses each semester.

Courses: Courses are identified by a unique course number and have a specific name.

- Each course can have multiple classes.

Classes : Each class pertains to a specific course and is identified by a section number.

Classes have a scheduled date and time and a record of how many students have registered.

- Classes are composed of multiple sections.

Requirements(2)

Sections: Each class section is identified by a unique section number.

Seat: Each class has multiple seats, each identified by a seat number and position within the classroom.

- Students fill these seats when they register for a class.

Instructors: Instructors are identified by a unique number and have a name and associated faculty.

- Each instructor can teach multiple courses.

Professors: Professors, like instructors, have a unique identifier, name, and faculty.

- Professors are responsible for teaching different sections of a class.

Tasks

- **Task 1:** Identify and define all necessary entities including **Students**, **Courses**, **Classes**, **Sections**, **Seats**, **Instructors**, and **Professors**. Assign appropriate attributes to each entity, particularly focusing on **primary keys** for unique identification.
- **Task 2:** Implement the relationships in your ERD, ensuring each relationship is clear and correctly labeled with the type of interaction (e.g., teaches, takes, fills).
- **Task 3:** Use an ERD tool or software to diagrammatically represent the university management system. Ensure the ERD is detailed, with all entities and relationships accurately represented with annotations.

Solution

